

# Implementation of Non-Standard Digital Signature Method in Vehicle-to-Everything Environments

*Master's Degree in Computer Engineering - Politecnico di Torino*

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## SUMMARY

Vehicle-to-Everything (V2X) technology is increasingly gaining traction worldwide and represents an important step towards improving driving safety by favoring communication among road users and infrastructures in real inter-vehicle networks. From a security perspective, V2X technology relies on a Public Key Infrastructure (PKI) and follows currently developed methods, particularly public-key algorithms for encrypting and signing messages. The security of public-key-based computer systems is at risk from the advent of quantum attacks. This presents a serious problem that systems, like V2X, must confront.

This thesis study is inserted in a set of researches conducted by Nardò Technical Center - Porsche Engineering, the Italian subsidiary of Porsche Engineering, which is active in various automotive domains for software development.

This thesis work focuses on a framework that enables the testing and evaluation of V2X communication integrated with Post-Quantum-Cryptography (PQC) algorithms, aiming to identify a point of confluence between these two largely relevant areas of study.

The specific objectives of the thesis are:

**1)** Integrating PQC APIs into the code used, to extend the standard followed for inter-vehicle communication. Specifically, this includes incorporating functions for key creation, message signing, and signature verification, which are crucial in the process of generating and sending CAM in a V2X environment.

**2)** Extending the structure and fields of the certificates of the entities involved in the V2X PKI infrastructure. Specifically, the RCA, AA, and AT use X.509 certificates where the keys are elliptic curve-based, as is the certificate signature. The expected outcome of this work is to generate new certificates that support the use of Post-Quantum Cryptography algorithms for the signing process.

3) Sending new non-standard CAM messages over the network interface, containing not only the new certificate inside the appropriate field of the Secured Header but also appending the new digital signature, generated using the specific PQC API and calculated on the unsecured part of the message.

4) Conducting a comparative analysis of message size and signing times between the existing standard method of sending CAM and the method implemented in this thesis.

In the development of the thesis, the process that led to choosing the framework used to integrate PQC algorithms into the V2X standards is highlighted. The challenges encountered and the decisions made to overcome them are described.

Initially, a proprietary solution was tested using Cohda Wireless Virtual Machines, which emulate the behavior of an On Board Unit (OBU) device. The original idea was to have a VM that works like the sender and another VM that works like a receiver, integrating the PQC APIs in the framework. The limitations encountered with the Cohda API set led the work to switch to an open-source solution.

The open-source tools ItsCertGen and FsMsgGen were selected. Respectively, the first is a testing tool used to produce certificates for V2X PKI entities starting from certificate profiles, and the second one uses the certificates produced by ItsCertGen to generate and inject CAM or DENM messages over the network interface. The code has been integrated with the library called liboqs, used for prototyping and testing with PQC APIs, and extended to support the new non-standard implementations.

The objective of replacing the current V2X standard algorithms for the digital signature process, based on elliptic curves, with a Post-Quantum algorithm is examined through the results of the developed implementations. A relative analysis demonstrates that the size of the newly produced messages is significantly increased rather to the standard ones. This huge difference in dimension could lead to various dangerous issues in the V2X context, in which the messages must be exchanged among entities in the order of milliseconds. A higher load of the communication channel, increased consumption of bandwidth leads to traffic congestion, reduction of message frequency, packet loss, and need for OBU with larger storage are different

examples of the consequences due to the analyzed result about message size. On the other side, the average times for signing messages and verifying signatures have been compared among the standard and non-standard solutions. The result shows comparable values between the chosen PQC algorithm and ECDSA, the current algorithm used in V2X communication.

In conclusion, this thesis study demonstrates how much the dimension of messages increases by integrating signatures and keys with longer dimensions, showing the actual unfeasibility of this implementation. However, it could be a starting point for a branch of research to face this issue. Thanks to the good feedback from the analysis of times, if the message size expansion is efficiently managed, a good balance for securing V2X environments against quantum attacks will be reached.